

Pablo Zúñiga Carvajal

Industrial & Systems Engineer, BSc & MSc Equivalent
Data Modeling, Applied Statistics & Complex Systems Thinking
Valparaíso, Chile / ESTA & H-1B1 Visa Eligible
pablo.zuniga.c@gmail.com / +56 9 4941 9529 / pablozunigac.github.io

Technical Skills

- **Enterprise Data Architecture & Ontologies:** ETL/ELT Pipelines, Metadata & Ontology Modeling, Data Governance, Enterprise Lineage, Relational Databases, Spatial Data Infrastructure (.gpx, .fit, .kmz), API REST
- **Code Languages & Advanced Analytics:** Python (Pandas, NumPy, SciPy), R (Quarto, Tidyverse, Tsibble), SQL, VBA, Statistical & Inferential Modeling, Time-Series Analysis, Linear & Integer Programming (LINGO, LINDO, MS Excel Solver/What'sBest!), Stochastic Simulation & Monte Carlo Methods (@RISK, Crystal Ball)
- **Infrastructure, Cloud & Orchestration:** Git/GitHub, Cloud Systems (MS Azure), Environment Orchestration, Distributed Protocols (Web3/On-Chain), Object-Oriented Programming (OOP)
- **Systems Architecture & Technical Tooling:** Complex Systems Modeling, BPMN / Process Engineering, Systems Diagramming (UML), Knowledge Architecture (Notion 2.0 Engines), Technical Documentation (Markdown, LaTeX, HTML/CSS)

Work Experience

Cuxhaven Labs | Principal Data & Systems Consultant | Remote **May 2026 – Present**
Architected production-grade ETL/ELT pipelines and metadata ontologies for critical infrastructure under strict auditability standards. Engineered operational resilience frameworks and disaster recovery protocols to guarantee continuous operational uptime during system incidents.

Independent Forward-Deployed Engineering Consultant | Remote **Jul 2022 – Present**
Telemetry Analytics Pipeline: Built an end-to-end time-series analytics pipeline in Python/R processing pace-distance telemetry across 1.8k+ athletes to quantify physiological gaps and optimize pacing strategies under multi-variable environmental conditions. Designed relational database structures, automated data flows, and technical documentation architectures using Git/Markdown.

Decentralized Systems & Protocols | On-Chain Operations Specialist | Remote **Jan 2020 – Jul 2022**
Managed multi-sig and non-custodial treasury assets (\$50k+) across Polkadot, Ethereum, and BSC networks. Executed dynamic gas optimization, cross-chain messaging, and smart contract utilities (1,100+ active users).

Banco de Crédito e Inversiones | Process Engineer (Data Governance) | Santiago, Chile **May 2019 – Sep 2019**
Orchestrated automated metadata integration pipelines from core transactional systems into MS Azure, standardizing BPMN workflows. Audited data architecture compliance in regulated banking environments, enforcing schema consistency and RBAC security controls.

ESVAL (Water Utility) | Lead Data & Project Engineer | Valparaíso, Chile **Jul 2017 – Dec 2017**
Developed a spatio-temporal risk model using Cox Proportional Hazards in R across 5,300 km of pipeline serving 1.9M+ residents. Delivered predictive maintenance risk scores via Agile/PMO frameworks to prioritize capital expenditure allocation for regional grid resilience.

ENAMI & Antofagasta Minerals | Industrial Engineering Intern | Chile **2014 & 2016**
Audited metallurgical inventory database (75k+ SKUs), uncovering 2.5M USD in accounting discrepancies for corporate financial audits. Monitored open-pit mining geotechnical telemetry for slope failure prediction and early risk mitigation in mission-critical environments.

Education

Pontificia Universidad Católica de Valparaíso | Valparaíso, Chile
Industrial & Systems Engineering (Integrated 6-Year Program – BSc & MSc Equivalent) | 2011 – 2017
Core Academic Foundations: Statistics, Systems Theory, Operations Research, Econometrics, Systems Modeling.

Leadership & Teaching

Graduate Teaching Assistant (PUCV)
Instructed Applied Data Science, MIS, and Quantitative Mathematics for 100+ engineering students (2018–2021).

Technical Instructor (BancoEstado / UTFSM)
Delivered time-series analytics and high-frequency financial modeling training to institutional trading desk staff.